

Alkali-Methyl Iodide Reactions

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Using empirical Monte Carlo trajectory procedures, we fitted a detailed 6-atom potential energy surface to molecular beam data for $\text{Rb} + \text{CH}_3\text{I} \rightarrow \text{RbI} + \text{CH}_3$. Measurements at a single energy of product recoil velocity and scattering angle distributions, and of the effect of aligning the CH_3I , were employed. We then used the potential surface to predict the energy dependence of the reactive cross-section, and to ascertain the consequences of omitting the CH_3 hydrogens from the calculations. Experimental data with which to compare our results are expected to appear.

Molecular beam studies of the alkali + CH_3I reactions have a long history of interaction with trajectory calculations. The first comparison of a Monte Carlo calculation¹ with experimental results² was made in 1962. Re-assessments of the situation have appeared at intervals.³⁻⁵ The sophistication of the experiments and of their interpretation have increased, and the trajectory procedures have become more general. There has been a steady growth in the reliability of the potential energy hypersurface that can be extracted from the experimental data by this empirical approach.

Recently there have been complementary further advances in the experiments and the calculations. The spatial orientation of CH_3I has become an experimental variable,⁶ and cross-sections have been measured as a function of reactant relative translational energy.⁷ In the trajectory calculations, the necessity for treating CH_3 as a single structureless particle disappeared when the first study of a CH_4 reaction was made.⁸ This makes possible a more detailed comparison than has been done before. We record here the current status of our efforts to provide one.

ETHOD

The technology of the calculation features of the present study, viz., method of selecting Monte Carlo symmetric top.

The potential energy is

s been reviewed.⁹ We describe the unique empirical potential function we used, and the conditions when one of the reactants is a

$$U = U_{\text{HMF}} + U_{3\text{H}} + U_{\text{R}}. \quad (1)$$

This was pieced together from several sources. U_{HMF} is a standard three-particle $\text{A} + \text{BC}$ type of function that has been fully described elsewhere.¹⁰ It provides the alkali-I-C part of the interaction. The $U_{3\text{H}}$ term represents the methyl hydrogens and their collapse to a planar configuration at the end of the reaction. It was taken from the first CH_4 calculation.⁸ There are three alkali-H repulsions in U_{R} , each of the form

$$D_{\text{R}} \exp [-\beta_{\text{R}}(r - r_{\text{R}})]. \quad (2)$$

All the potential parameters are identified in table 1. Bonding parameters for stable

reactant and product molecules are also present, and can be identified from the listed references.

For starting conditions, we begin with a randomly oriented CH_3I molecule. (This involves three random rotations from an initially fixed position, if a general molecular beam experiment is being simulated. For experiments with oriented

TABLE 1.—PARAMETERS OF THE EMPIRICAL POTENTIAL

parameter	purpose	ref.	final value
REACTION PATH : (U_{HMF}) :			
u_0	sharpness of corner	10, 3	$1/9 \text{ \AA}^2$
v_0	location of downhill	10, 3	$-1 \text{ \AA}^2$
a	steepness of downhill	10, 3	$4 \text{ \AA}^{-2}$
ENTRANCE VALLEY CONSTRICTION (U_{HMF}) :			
v^*	location	10	$-4.5 \text{ \AA}^2$
L^*	characteristic v^*	10	$9 \text{ \AA}^{-2}$
β^*	width	10	$6 \text{ \AA}^{-1}$
ANGLE-DEPENDENT FORCES (U_{HMF}) :			
n	directivity	3	1
d	intensity	3	23.9 kcal
UMBRELLA FUNCTION ($U_{3\text{H}}$) :			
r_v	CH_3 collapse rate	8	$2.7 \text{ \AA}$
r_β	softness	8	$4.2 \text{ \AA}$
ALKALI-H REPULSIONS (U_{R}) :			
D_{R}	intensity	eqn. (2)	100 kcal
β_{R}	suddenness	eqn (2)	$4 \text{ \AA}^{-1}$
r_{R}	characteristic distance	eqn (2)	$2.8 \text{ \AA}$

CH_3I we omit two of them. Simulation of this case is for extrapolated perfect alignment. The alignment axis is z .) The CH_3I angular momentum and its z component are sampled from the probability distributions ¹¹

$$P(L_z) = \exp(-L_z^2/2I_zkT), \quad 0 \leq L_z \leq \infty; \quad (3)$$

$$P(L) = L \exp(-L^2/2I_xkT), \quad L_z \leq L \leq \infty. \quad (4)$$

Eqn (3) can be sampled by rejection, and eqn (4) by the cdf formula

$$L = [L_z^2 - 2I_xkT \ln(1-R)]^{1/2}, \quad (5)$$

R being a random number $0 \leq R \leq 1$. The vector

$$\omega = (L_z/I_z)\mathbf{e}_z + (L \sin \alpha \sin \beta/I_x)\mathbf{e}_x + (L \sin \alpha \cos \beta/I_x)\mathbf{e}_y, \quad (6)$$

with $\cos \alpha = L_z/L$, β random $0 \leq \beta \leq 2\pi$, and $\mathbf{e}$ unit vectors, is subjected to the same rotation as CH_3I was. The rotational atomic momentum contributions are then given by the usual $\mathbf{p}_i = m_i \omega \wedge \mathbf{r}_i$. (For oriented molecule experiments $L = L_z$ and most of this is unnecessary.)

The C—I bond has initial vibrational momentum,

$$p_v = S(-2kT\mu \ln R)^{1/2}, \quad (7)$$

with S a random sign and μ the reduced mass of I and CH_3 . There is also an initial displacement,

$$\beta^{-1} \ln [1 + (\rho_v^2/2\mu D)^{1/2}] \sin [\pi(R - \frac{1}{2})], \quad (8)$$

obtained by sampling the classical phase distribution between turning points, with β and D Morse parameters for the C—I bond. These contributions are distributed vectorially over the molecule in the way appropriate to its actual orientation. Lastly, the impact parameter and alkali velocity vector orientation are chosen by standard methods.

RESULTS

Our results take the form of a tentative potential surface for $\text{Rb} + \text{CH}_3\text{I}$, based on molecular beam evidence at fixed reactant energy, plus a prediction of the, as yet, unmeasured energy-dependent cross-section for reactive scattering. The reason for this is that although the cross-sections as a function of energy have been measured for the K reaction, the corresponding aligned- CH_3I data are insufficiently precise for our purposes.

The final parameters for $\text{Rb} + \text{CH}_3\text{I}$ are given in table 1. We determined these by requiring that the RbI scattering be peaked into the backward hemisphere, the product recoil velocity be as large as possible, and the cross-section ratio for extrapolated perfect alignment/anti-alignment be about 4.^{6,7} The peak scattering angle for the surface finally selected was 110° , the average recoil energy was a little over

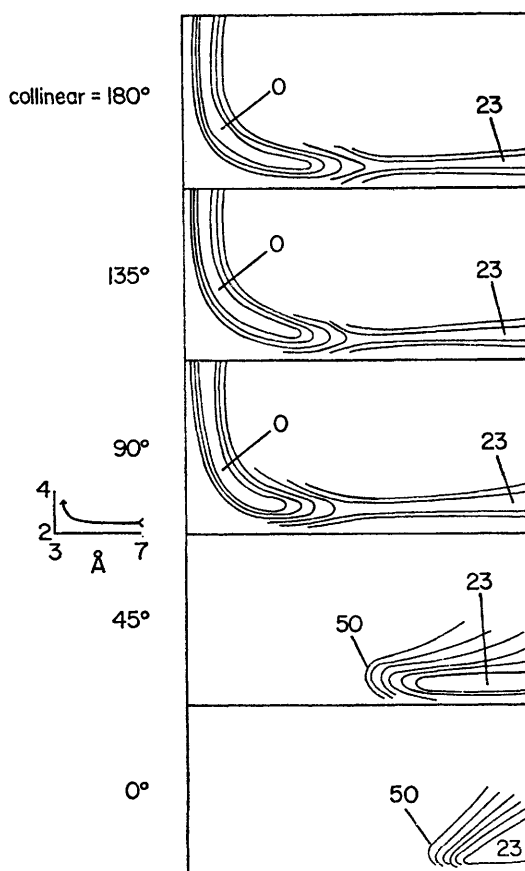


FIG. 1.—Potential energy surface for $\text{Rb} + \text{CH}_3\text{I}$, with several angles of approach. Contour intervals are 5 kcal, descending from lower right to upper left for 180° , 135° and 90° ascending for the others.

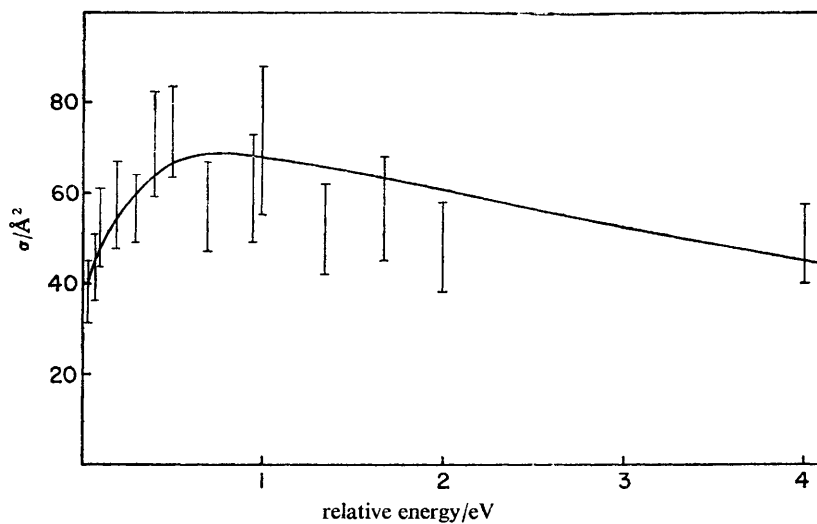


FIG. 2.—Predicted cross-section as a function of relative energy for $\text{Rb} + \text{CH}_3\text{I}$.

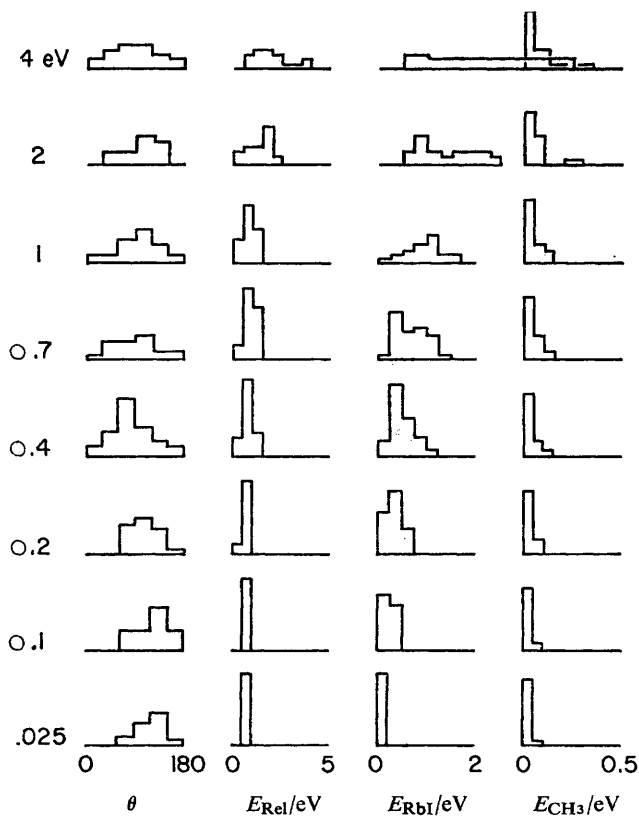


FIG. 3.—Distributions of scattering angle θ , product recoil energy E_{Rel} , RbI internal energy (E_{RbI} , vibration and rotation) and CH_3 internal energy (E_{CH_3}) at various reactant energies. $\theta = 0$ corresponds to forward scattering.

0.8 of exoergicity with a perceptible tail towards lower values, and the alignment/anti-alignment reactivity ratio was 3 ± 1 .

It was not easy to attain this result. We rejected 32 potential surfaces in the course of reaching it. (A list of their parameters and individual defects can be supplied, but its presentation here would not be illustrative, because the logical sequence was disrupted by changes in the experimenters' interpretation of their data. The ranges of parameter variation were $v_0 = \pm 1$, $u_0 = \frac{1}{2}$ to $1/100$, $a = 1$ to 6 , $n = 1$ to 4 , $d = 0$ to 47.8 , $\beta^* = 3$ to 8 .) To obtain the scattering and alignment/anti-alignment ratio simultaneously correct involves a delicate combination of valence-type forces (n , d) and Rb—H repulsions. At the same time the cross-section has to be kept low (a , β^*) and the recoil velocity high (v_0 , mostly). The parameter variations interact with one another to a much larger extent than has been found in 3-particle calculations.

The potential surface has been plotted in fig. 1. The plots are for collinear Rb approach, and for Rb—I—C angles of 135° , 90° , 45° and 0° . The difference between the eclipsed and staggered configurations of Rb—I—C—H is perceptible on a large-scale map, but not on a figure of the size printed here.

The predicted reactive cross-section as a function of reactant relative translational energy is displayed in fig. 2. The point at 0.1 eV corresponds to the data used to establish the potential surface. The curve shown is arbitrary and is provided only as a guide.

Fig. 3 shows how the scattering and the energy disposal appear to change as the reactant energy increases. There is a slow shift from backward to sideways scattering—the reaction would probably approach the stripping regime at high enough energy, but 4 eV is not enough for this. Although most of the exoergicity appears as product recoil at low incident energy, the main effect of increasing reactant translation seems to be augmentation of the RbI internal energy. Product relative energy increases slowly, and CH_3 internal energy hardly at all. At 4 eV reactant approach energy, vibrationally cold CH_3 is still being produced.

COMPARISON WITH 3-PARTICLE CALCULATIONS

At 0.5 and 4 eV relative energy, we made calculations with the methyl hydrogens "suppressed". This was done by setting the Rb—H repulsions and all the C—H force constants to zero, making the hydrogens into free particles that dispersed without interacting with the rest of the system. The calculation is for comparison with

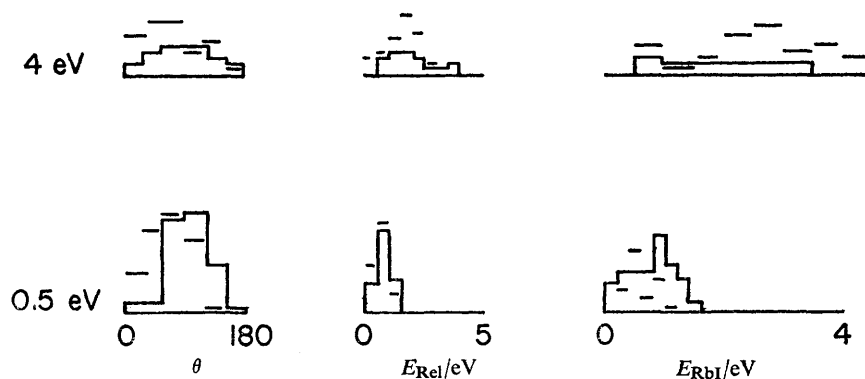


FIG. 4.—Effect of suppressing the methyl hydrogens. The solid histograms are similar to those in fig. 3. The broken ones are with the hydrogens suppressed.

the 3-particle study recently made by LaBudde, Kuntz, Bernstein, and Levine.¹² They treated K rather than Rb. Using K in our calculation leads to essentially the same comparative results as are exhibited here for Rb.

In fig. 4 we show the effect of hydrogen suppression (broken histograms) on the kind of data displayed in fig. 3. There is a moderately small forward shift in the scattering, and at high energies an increased internal energization of RbI. The other data are unaffected, within the noise level of the calculation. The cross-sections are also about the same when the hydrogens are removed, but—and most importantly—alignment discrimination is lost, and the unfavourably aligned CH₃I are as reactive as the favourably aligned ones. This last item is from an additional independent calculation at 0.1 eV relative energy.

DISCUSSION

Our results must be considered provisional. We expect to have to refine our potential surface further, if the anticipated (cross-section, energy) measurements for Rb+CH₃I do not support our prediction. To this end, we have kept the resolution of our calculations minimal, to provide a reserve if it is needed.

We have published¹³ our calculation of the K+CH₃I cross-section as a function of energy, on the assumption that the K and Rb potential surfaces are similar. (This assumption is demonstrably incorrect, at least as regards the ability of the same angle-dependent forces to provide reasonable alignment/anti-alignment reactivity ratios for both reactions.) Our cross-sections for K peak where the experimental ones do, but do not decrease nearly as rapidly with increasing energy. It remains to be seen whether the Rb comparison will also have this feature. If it does, and if further manipulation of the potential parameters does not reproduce it, we may have to consider allowing transitions between ionic and covalent surfaces to play a role in the trajectories. We have rejected so many surfaces in the course of the work reported here that our remaining flexibility of choice cannot be large.

The 3-particle calculation of LaBudde, Kuntz, Bernstein and Levine¹² employs the locally lower of two potential surfaces, ionic and covalent, with the crossing near the closest approach of all the atoms (roughly, $v_0 = 0$ in our terms). Our potential also has a sharp transition (large a), but this arose naturally from the experimental data rather than by advance specification. Our crossing—if that is indeed what it corresponds to—is much earlier in the reactant channel than is true in the 3-particle representations. However, before we began trying to encompass the aligned-CH₃I data, our transition points also tended to be near $v_0 = 0$.

It is too early to assess the relationship between these two trajectory calculations. The 3-particle calculation is much less empirical than the one described here. Comparison is mainly with the cross-section energy dependence, and the other data take second priority. Our work does just the opposite. The aligned-CH₃I data are of crucial importance for us and have not been used in the 3-particle procedure, probably for the reasons given earlier. The K—Rb mismatch of experimental subject matter continues to be a nuisance. But there is hope for clarification, possibly by the time this paper is published.

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